

CSS Roadmap (Beginner to Advanced)

1. CSS Basics

- What is CSS (Cascading Style Sheets)?
- Inline, Internal, and External CSS
- Selectors: element, class (.class), ID (#id)
- Basic properties:
 - color, background-color
 - font-size, font-family
 - width, height, margin, padding, border
- CSS Comments /* comment */

Practice: Style a simple HTML page with headings and paragraphs.

2. Intermediate CSS

- Box Model (content + padding + border + margin)
- display: block, inline, inline-block, none
- Positioning: static, relative, absolute, fixed, sticky
- z-index and stacking context
- Units: px, em, rem, %, vh, vw
- Pseudo-classes: :hover, :focus, :nth-child()
- Pseudo-elements: ::before, ::after
- opacity, visibility, overflow
- cursor, text-align, vertical-align

Practice idea: Create a pricing card layout.

3. Advanced Layout Techniques

- Flexbox:
 - display: flex
 - justify-content, align-items, flex-direction, gap

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- Grid:
 - display: grid
 - grid-template-columns, grid-gap, grid-area
- media queries for responsive design
- min-width, max-width, clamp(), calc()
- Responsive Design Principles
- CSS Variables (--main-color)
- aspect-ratio for images and videos

Project idea: Responsive navbar, multi-column layout, mobile-friendly landing page.

4. CSS Animations & Transitions

- transition: property, duration, timing-function
- transform: scale(), rotate(), translate()
- animation: @keyframes, animation-name, animation-duration
- CSS :checked & checkbox hacks
- will-change and performance tips

Project: Animated buttons, loader spinner, accordion, modal popup.

5. Modern CSS Tools & Architecture

- BEM Naming Convention
- DRY CSS - Don't Repeat Yourself
- SCSS/SASS Basics: Variables, nesting, mixins, functions, partials
- Tailwind CSS (Utility-first CSS framework)
- CSS Resets & Normalize.css
- CSS Preprocessors (SASS, PostCSS)

6. Master-Level Techniques

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- CSS Custom Properties (Dynamic theming)
- Dark Mode using CSS (prefers-color-scheme)
- Print Stylesheets
- Logical properties: margin-inline, padding-block, etc.
- Accessibility (focus styles, contrast, reduced motion)
- Performance Optimization: Minification, Critical CSS
- Browser compatibility tools (Can I use, Autoprefixer)

7. Practice Projects

- Portfolio Website (with Flexbox/Grid)
- Blog Layout
- Image Gallery
- Responsive Form UI
- Product Card with Hover Effects
- Landing Page with Animations
- Dark/Light Toggle Mode

Bonus Tips

- Use DevTools to inspect and test CSS in real-time
- Bookmark <https://css-tricks.com/> and <https://developer.mozilla.org/en-US/docs/Web/CSS>
- Practice on Frontend Mentor, CodePen, or CSSBattle